```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df=pd.read_csv('Breast Cancer.csv')

df.shape
```

```
(569, 32)
```

```python
df.head()
```

```
         id  radius_mean  texture_mean  perimeter_mean  area_mean  \
0    842302        17.99         10.38          122.80     1001.0
1    842517        20.57         17.77          132.90     1326.0
2  84300903        19.69         21.25          130.00     1203.0
3  84348301        11.42         20.38           77.58      386.1
4  84358402        20.29         14.34          135.10     1297.0

   smoothness_mean  compactness_mean  concavity_mean  concave
points_mean  \
0          0.11840           0.27760          0.3001
0.14710
1          0.08474           0.07864          0.0869
0.07017
2          0.10960           0.15990          0.1974
0.12790
3          0.14250           0.28390          0.2414
0.10520
4          0.10030           0.13280          0.1980
0.10430

   symmetry_mean  ...  texture_worst  perimeter_worst  area_worst  \
0         0.2419  ...          17.33           184.60      2019.0
1         0.1812  ...          23.41           158.80      1956.0
2         0.2069  ...          25.53           152.50      1709.0
3         0.2597  ...          26.50            98.87       567.7
4         0.1809  ...          16.67           152.20      1575.0

   smoothness_worst  compactness_worst  concavity_worst  concave
points_worst  \
0            0.1622             0.6656           0.7119
0.2654
1            0.1238             0.1866           0.2416
0.1860
2            0.1444             0.4245           0.4504
0.2430
3            0.2098             0.8663           0.6869
0.2575
4            0.1374             0.2050           0.4000
```

```
0.1625

   symmetry_worst  fractal_dimension_worst  diagnosis
0          0.4601                  0.11890          M
1          0.2750                  0.08902          M
2          0.3613                  0.08758          M
3          0.6638                  0.17300          M
4          0.2364                  0.07678          M

[5 rows x 32 columns]
```

```
df.columns
```

```
Index(['id', 'radius_mean', 'texture_mean', 'perimeter_mean',
'area_mean',
       'smoothness_mean', 'compactness_mean', 'concavity_mean',
       'concave points_mean', 'symmetry_mean',
'fractal_dimension_mean',
       'radius_se', 'texture_se', 'perimeter_se', 'area_se',
'smoothness_se',
       'compactness_se', 'concavity_se', 'concave points_se',
'symmetry_se',
       'fractal_dimension_se', 'radius_worst', 'texture_worst',
       'perimeter_worst', 'area_worst', 'smoothness_worst',
       'compactness_worst', 'concavity_worst', 'concave points_worst',
       'symmetry_worst', 'fractal_dimension_worst', 'diagnosis'],
      dtype='object')
```

```
df.tail()
```

```
         id  radius_mean  texture_mean  perimeter_mean  area_mean  \
564  926424        21.56         22.39          142.00     1479.0
565  926682        20.13         28.25          131.20     1261.0
566  926954        16.60         28.08          108.30      858.1
567  927241        20.60         29.33          140.10     1265.0
568   92751         7.76         24.54           47.92      181.0

     smoothness_mean  compactness_mean  concavity_mean  concave
points_mean  \
564          0.11100           0.11590         0.24390
0.13890
565          0.09780           0.10340         0.14400
0.09791
566          0.08455           0.10230         0.09251
0.05302
567          0.11780           0.27700         0.35140
0.15200
568          0.05263           0.04362         0.00000
0.00000

     symmetry_mean  ...  texture_worst  perimeter_worst  area_worst  \
```

```
564         0.1726  ...          26.40           166.10      2027.0
565         0.1752  ...          38.25           155.00      1731.0
566         0.1590  ...          34.12           126.70      1124.0
567         0.2397  ...          39.42           184.60      1821.0
568         0.1587  ...          30.37            59.16       268.6

     smoothness_worst  compactness_worst  concavity_worst  \
564           0.14100            0.21130           0.4107
565           0.11660            0.19220           0.3215
566           0.11390            0.30940           0.3403
567           0.16500            0.86810           0.9387
568           0.08996            0.06444           0.0000

     concave points_worst  symmetry_worst  fractal_dimension_worst
diagnosis
564                0.2216          0.2060                  0.07115
M
565                0.1628          0.2572                  0.06637
M
566                0.1418          0.2218                  0.07820
M
567                0.2650          0.4087                  0.12400
M
568                0.0000          0.2871                  0.07039
B

[5 rows x 32 columns]
```

```
df.sample(5)
```

```
          id  radius_mean  texture_mean  perimeter_mean  area_mean  \
550   923748        10.86         21.48           68.51      360.5
42    855625        19.07         24.81          128.30     1104.0
165  8712291        14.97         19.76           95.50      690.2
326    89524        14.11         12.88           90.03      616.5
54    857438        15.10         22.02           97.26      712.8

     smoothness_mean  compactness_mean  concavity_mean  concave
points_mean  \
550          0.07431           0.04227         0.00000
0.00000
42           0.09081           0.21900         0.21070
0.09961
165          0.08421           0.05352         0.01947
0.01939
326          0.09309           0.05306         0.01765
0.02733
54           0.09056           0.07081         0.05253
0.03334
```

```
     symmetry_mean  ...  texture_worst  perimeter_worst  area_worst  \
550         0.1661  ...          24.77            74.08       412.3
42          0.2310  ...          33.17           177.40      1651.0
165         0.1515  ...          25.82           102.30       782.1
326         0.1373  ...          18.00            98.40       749.9
54          0.1616  ...          31.69           117.70      1030.0

     smoothness_worst  compactness_worst  concavity_worst  \
550            0.1001            0.07348          0.00000
42             0.1247            0.74440          0.72420
165            0.1045            0.09995          0.07750
326            0.1281            0.11090          0.05307
54             0.1389            0.20570          0.27120

     concave points_worst  symmetry_worst  fractal_dimension_worst
diagnosis
550               0.00000          0.2458                  0.06592
B
42                0.24930          0.4670                  0.10380
M
165               0.05754          0.2646                  0.06085
B
326               0.05890          0.2100                  0.07083
B
54                0.15300          0.2675                  0.07873
M

[5 rows x 32 columns]
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 569 entries, 0 to 568
Data columns (total 32 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   id                       569 non-null    int64
 1   radius_mean              569 non-null    float64
 2   texture_mean             569 non-null    float64
 3   perimeter_mean           569 non-null    float64
 4   area_mean                569 non-null    float64
 5   smoothness_mean          569 non-null    float64
 6   compactness_mean         569 non-null    float64
 7   concavity_mean           569 non-null    float64
 8   concave points_mean      569 non-null    float64
 9   symmetry_mean            569 non-null    float64
 10  fractal_dimension_mean   569 non-null    float64
 11  radius_se                569 non-null    float64
 12  texture_se               569 non-null    float64
 13  perimeter_se             569 non-null    float64
```

```
 14  area_se                   569 non-null    float64
 15  smoothness_se             569 non-null    float64
 16  compactness_se            569 non-null    float64
 17  concavity_se              569 non-null    float64
 18  concave points_se         569 non-null    float64
 19  symmetry_se               569 non-null    float64
 20  fractal_dimension_se      569 non-null    float64
 21  radius_worst              569 non-null    float64
 22  texture_worst             569 non-null    float64
 23  perimeter_worst           569 non-null    float64
 24  area_worst                569 non-null    float64
 25  smoothness_worst          569 non-null    float64
 26  compactness_worst         569 non-null    float64
 27  concavity_worst           569 non-null    float64
 28  concave points_worst      569 non-null    float64
 29  symmetry_worst            569 non-null    float64
 30  fractal_dimension_worst   569 non-null    float64
 31  diagnosis                 569 non-null    object
dtypes: float64(30), int64(1), object(1)
memory usage: 142.4+ KB
```

```
df.dtypes
```

```
id                           int64
radius_mean                float64
texture_mean               float64
perimeter_mean             float64
area_mean                  float64
smoothness_mean            float64
compactness_mean           float64
concavity_mean             float64
concave points_mean        float64
symmetry_mean              float64
fractal_dimension_mean     float64
radius_se                  float64
texture_se                 float64
perimeter_se               float64
area_se                    float64
smoothness_se              float64
compactness_se             float64
concavity_se               float64
concave points_se          float64
symmetry_se                float64
fractal_dimension_se       float64
radius_worst               float64
texture_worst              float64
perimeter_worst            float64
area_worst                 float64
smoothness_worst           float64
compactness_worst          float64
```

```
concavity_worst             float64
concave points_worst        float64
symmetry_worst              float64
fractal_dimension_worst     float64
diagnosis                    object
dtype: object
```

```
df.describe().T
```

```
df.isnull().sum()
```

```
id                          0
radius_mean                 0
texture_mean                0
perimeter_mean              0
area_mean                   0
smoothness_mean             0
compactness_mean            0
concavity_mean              0
concave points_mean         0
symmetry_mean               0
fractal_dimension_mean      0
radius_se                   0
texture_se                  0
perimeter_se                0
area_se                     0
smoothness_se               0
compactness_se              0
concavity_se                0
concave points_se           0
symmetry_se                 0
fractal_dimension_se        0
radius_worst                0
texture_worst               0
perimeter_worst             0
area_worst                  0
smoothness_worst            0
compactness_worst           0
concavity_worst             0
concave points_worst        0
symmetry_worst              0
fractal_dimension_worst     0
diagnosis                   0
dtype: int64
```

```
df.duplicated()
```

```
0      False
1      False
2      False
```

```
3      False
4      False
       ...
564    False
565    False
566    False
567    False
568    False
Length: 569, dtype: bool
```

```
df.nunique()
```

```
id                           569
radius_mean                  456
texture_mean                 479
perimeter_mean               522
area_mean                    539
smoothness_mean              474
compactness_mean             537
concavity_mean               537
concave points_mean          542
symmetry_mean                432
fractal_dimension_mean       499
radius_se                    540
texture_se                   519
perimeter_se                 533
area_se                      528
smoothness_se                547
compactness_se               541
concavity_se                 533
concave points_se            507
symmetry_se                  498
fractal_dimension_se         545
radius_worst                 457
texture_worst                511
perimeter_worst              514
area_worst                   544
smoothness_worst             411
compactness_worst            529
concavity_worst              539
concave points_worst         492
symmetry_worst               500
fractal_dimension_worst      535
diagnosis                      2
dtype: int64
```

```
df.drop('id',axis=1,inplace=True)
```

```
df.head()
```

```
   radius_mean  texture_mean  perimeter_mean  area_mean
smoothness_mean  \
0        17.99         10.38          122.80     1001.0
0.11840
1        20.57         17.77          132.90     1326.0
0.08474
2        19.69         21.25          130.00     1203.0
0.10960
3        11.42         20.38           77.58      386.1
0.14250
4        20.29         14.34          135.10     1297.0
0.10030

   compactness_mean  concavity_mean  concave points_mean
symmetry_mean  \
0           0.27760          0.3001              0.14710
0.2419
1           0.07864          0.0869              0.07017
0.1812
2           0.15990          0.1974              0.12790
0.2069
3           0.28390          0.2414              0.10520
0.2597
4           0.13280          0.1980              0.10430
0.1809

   fractal_dimension_mean  ...  texture_worst  perimeter_worst
area_worst  \
0                 0.07871  ...          17.33           184.60
2019.0
1                 0.05667  ...          23.41           158.80
1956.0
2                 0.05999  ...          25.53           152.50
1709.0
3                 0.09744  ...          26.50            98.87
567.7
4                 0.05883  ...          16.67           152.20
1575.0

   smoothness_worst  compactness_worst  concavity_worst  concave
points_worst  \
0            0.1622             0.6656           0.7119
0.2654
1            0.1238             0.1866           0.2416
0.1860
2            0.1444             0.4245           0.4504
0.2430
3            0.2098             0.8663           0.6869
0.2575
4            0.1374             0.2050           0.4000
```

```
0.1625

   symmetry_worst  fractal_dimension_worst  diagnosis
0          0.4601                  0.11890          M
1          0.2750                  0.08902          M
2          0.3613                  0.08758          M
3          0.6638                  0.17300          M
4          0.2364                  0.07678          M

[5 rows x 31 columns]
```

```
df.shape
```

```
(569, 31)
```

```
df.columns
```

```
Index(['radius_mean', 'texture_mean', 'perimeter_mean', 'area_mean',
       'smoothness_mean', 'compactness_mean', 'concavity_mean',
       'concave points_mean', 'symmetry_mean',
'fractal_dimension_mean',
       'radius_se', 'texture_se', 'perimeter_se', 'area_se',
'smoothness_se',
       'compactness_se', 'concavity_se', 'concave points_se',
'symmetry_se',
       'fractal_dimension_se', 'radius_worst', 'texture_worst',
       'perimeter_worst', 'area_worst', 'smoothness_worst',
       'compactness_worst', 'concavity_worst', 'concave points_worst',
       'symmetry_worst', 'fractal_dimension_worst', 'diagnosis'],
      dtype='object')
```

```
df['diagnosis'] = df['diagnosis'].map({'M': 0, 'B': 1})
```

```
df.head()
```

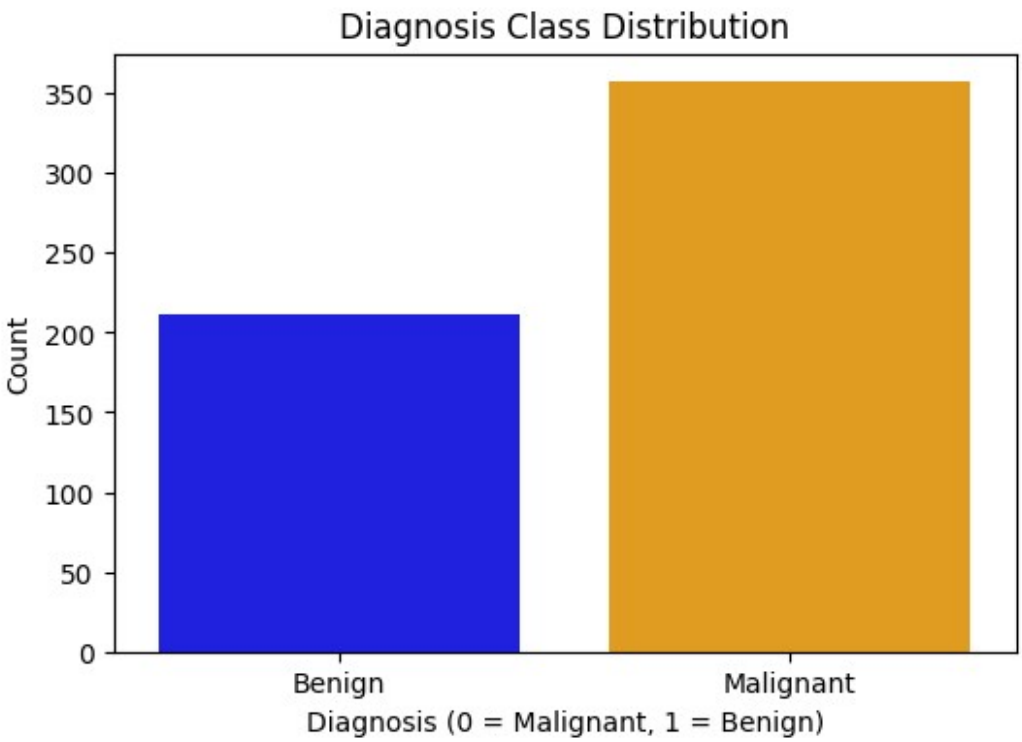
```
   radius_mean  texture_mean  perimeter_mean  area_mean
smoothness_mean  \
0        17.99         10.38          122.80     1001.0
0.11840
1        20.57         17.77          132.90     1326.0
0.08474
2        19.69         21.25          130.00     1203.0
0.10960
3        11.42         20.38           77.58      386.1
0.14250
4        20.29         14.34          135.10     1297.0
0.10030

   compactness_mean  concavity_mean  concave points_mean
symmetry_mean  \
0           0.27760          0.3001              0.14710
```

```
0.2419
1           0.07864          0.0869              0.07017
0.1812
2           0.15990          0.1974              0.12790
0.2069
3           0.28390          0.2414              0.10520
0.2597
4           0.13280          0.1980              0.10430
0.1809

   fractal_dimension_mean  ...  texture_worst  perimeter_worst
area_worst  \
0                 0.07871  ...          17.33           184.60
2019.0
1                 0.05667  ...          23.41           158.80
1956.0
2                 0.05999  ...          25.53           152.50
1709.0
3                 0.09744  ...          26.50            98.87
567.7
4                 0.05883  ...          16.67           152.20
1575.0

   smoothness_worst  compactness_worst  concavity_worst  concave
points_worst  \
0            0.1622             0.6656           0.7119
0.2654
1            0.1238             0.1866           0.2416
0.1860
2            0.1444             0.4245           0.4504
0.2430
3            0.2098             0.8663           0.6869
0.2575
4            0.1374             0.2050           0.4000
0.1625

   symmetry_worst  fractal_dimension_worst  diagnosis
0          0.4601                  0.11890          0
1          0.2750                  0.08902          0
2          0.3613                  0.08758          0
3          0.6638                  0.17300          0
4          0.2364                  0.07678          0

[5 rows x 31 columns]
```

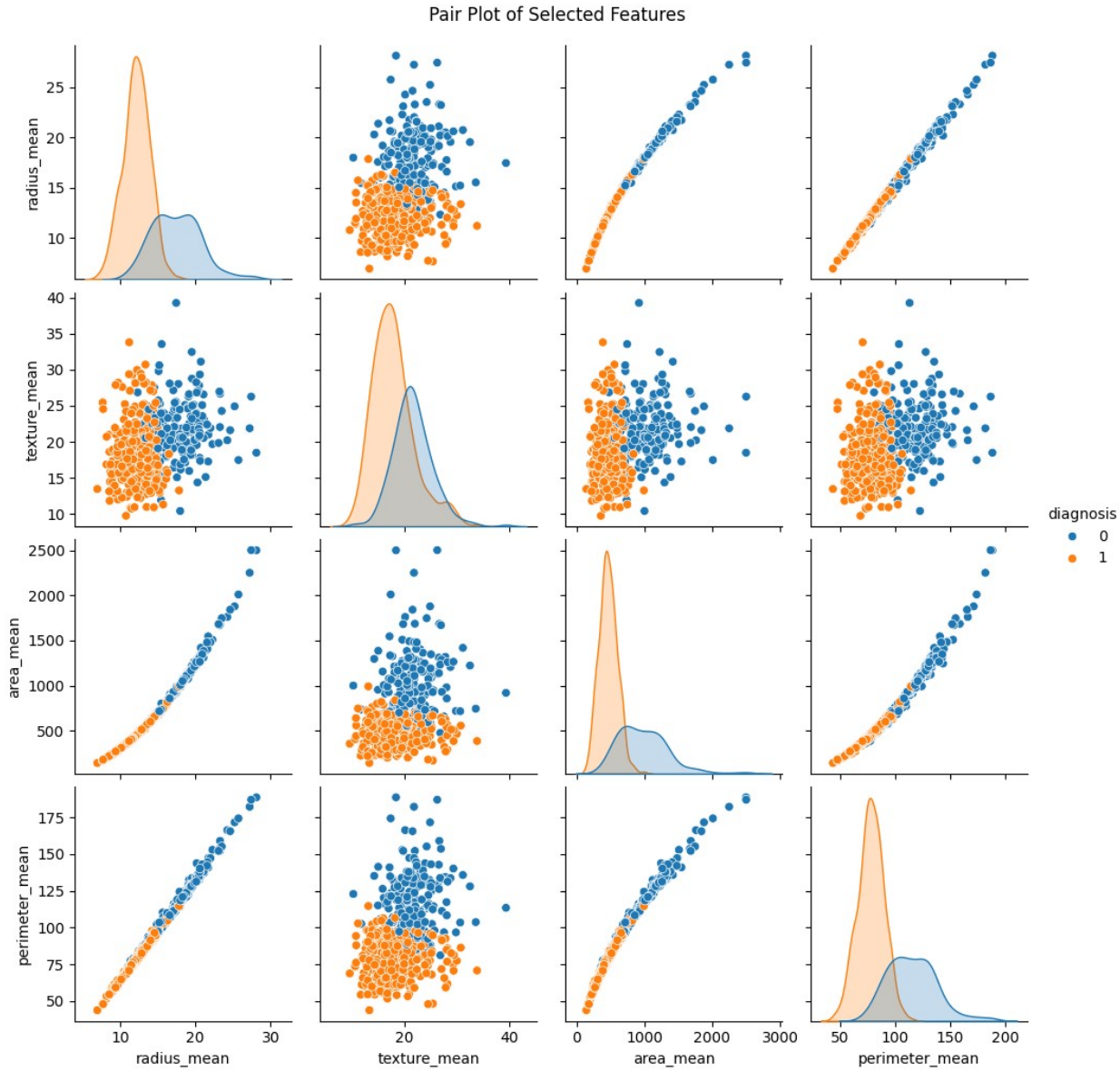
```
#Countplot
plt.figure(figsize=(6,4))
sns.countplot(x=df['diagnosis'],palette=["blue","orange"])
plt.xticks(ticks=[0,1],labels=['Benign','Malignant'])
plt.title('Diagnosis Class Distribution')
```

```
plt.xlabel('Diagnosis (0 = Malignant, 1 = Benign)')
plt.ylabel('Count')
plt.show()
```

```
C:\Users\LENOVO\AppData\Local\Temp\ipykernel_9776\2421332341.py:2:
FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be
removed in v0.14.0. Assign the `x` variable to `hue` and set
`legend=False` for the same effect.

  sns.countplot(x=df['diagnosis'],palette=["blue","orange"])
```




```
#Pairplot
sns.pairplot(df[['radius_mean', 'texture_mean', 'area_mean',
'perimeter_mean', 'diagnosis']], hue='diagnosis')
plt.suptitle('Pair Plot of Selected Features', y=1.02)
plt.show()
```




```
df.corr()
```

```
                          radius_mean  texture_mean  perimeter_mean
area_mean  \
radius_mean                  1.000000      0.323782        0.997855
0.987357
texture_mean                 0.323782      1.000000        0.329533
0.321086
perimeter_mean               0.997855      0.329533        1.000000
0.986507
area_mean                    0.987357      0.321086        0.986507
1.000000
smoothness_mean              0.170581     -0.023389        0.207278
0.177028
```

compactness_mean	0.506124	0.236702	0.556936	0.498502
concavity_mean	0.676764	0.302418	0.716136	0.685983
concave points_mean	0.822529	0.293464	0.850977	0.823269
symmetry_mean	0.147741	0.071401	0.183027	0.151293
fractal_dimension_mean	-0.311631	-0.076437	-0.261477	-0.283110
radius_se	0.679090	0.275869	0.691765	0.732562
texture_se	-0.097317	0.386358	-0.086761	-0.066280
perimeter_se	0.674172	0.281673	0.693135	0.726628
area_se	0.735864	0.259845	0.744983	0.800086
smoothness_se	-0.222600	0.006614	-0.202694	-0.166777
compactness_se	0.206000	0.191975	0.250744	0.212583
concavity_se	0.194204	0.143293	0.228082	0.207660
concave points_se	0.376169	0.163851	0.407217	0.372320
symmetry_se	-0.104321	0.009127	-0.081629	-0.072497
fractal_dimension_se	-0.042641	0.054458	-0.005523	-0.019887
radius_worst	0.969539	0.352573	0.969476	0.962746
texture_worst	0.297008	0.912045	0.303038	0.287489
perimeter_worst	0.965137	0.358040	0.970387	0.959120
area_worst	0.941082	0.343546	0.941550	0.959213
smoothness_worst	0.119616	0.077503	0.150549	0.123523
compactness_worst	0.413463	0.277830	0.455774	0.390410
concavity_worst	0.526911	0.301025	0.563879	0.512606
concave points_worst	0.744214	0.295316	0.771241	0.722017
symmetry_worst	0.163953	0.105008	0.189115	0.143570
fractal_dimension_worst	0.007066	0.119205	0.051019	

```
0.003738
diagnosis                   -0.730029     -0.415185       -0.742636  -
0.708984

                          smoothness_mean  compactness_mean
concavity_mean  \
radius_mean                      0.170581          0.506124
0.676764
texture_mean                    -0.023389          0.236702
0.302418
perimeter_mean                   0.207278          0.556936
0.716136
area_mean                        0.177028          0.498502
0.685983
smoothness_mean                  1.000000          0.659123
0.521984
compactness_mean                 0.659123          1.000000
0.883121
concavity_mean                   0.521984          0.883121
1.000000
concave points_mean              0.553695          0.831135
0.921391
symmetry_mean                    0.557775          0.602641
0.500667
fractal_dimension_mean           0.584792          0.565369
0.336783
radius_se                        0.301467          0.497473
0.631925
texture_se                       0.068406          0.046205
0.076218
perimeter_se                     0.296092          0.548905
0.660391
area_se                          0.246552          0.455653
0.617427
smoothness_se                    0.332375          0.135299
0.098564
compactness_se                   0.318943          0.738722
0.670279
concavity_se                     0.248396          0.570517
0.691270
concave points_se                0.380676          0.642262
0.683260
symmetry_se                      0.200774          0.229977
0.178009
fractal_dimension_se             0.283607          0.507318
0.449301
radius_worst                     0.213120          0.535315
0.688236
texture_worst                    0.036072          0.248133
```

```
0.299879
perimeter_worst                  0.238853          0.590210
0.729565
area_worst                       0.206718          0.509604
0.675987
smoothness_worst                 0.805324          0.565541
0.448822
compactness_worst                0.472468          0.865809
0.754968
concavity_worst                  0.434926          0.816275
0.884103
concave points_worst             0.503053          0.815573
0.861323
symmetry_worst                   0.394309          0.510223
0.409464
fractal_dimension_worst          0.499316          0.687382
0.514930
diagnosis                       -0.358560         -0.596534        -
0.696360

                          concave points_mean  symmetry_mean  \
radius_mean                          0.822529       0.147741
texture_mean                         0.293464       0.071401
perimeter_mean                       0.850977       0.183027
area_mean                            0.823269       0.151293
smoothness_mean                      0.553695       0.557775
compactness_mean                     0.831135       0.602641
concavity_mean                       0.921391       0.500667
concave points_mean                  1.000000       0.462497
symmetry_mean                        0.462497       1.000000
fractal_dimension_mean               0.166917       0.479921
radius_se                            0.698050       0.303379
texture_se                           0.021480       0.128053
perimeter_se                         0.710650       0.313893
area_se                              0.690299       0.223970
smoothness_se                        0.027653       0.187321
compactness_se                       0.490424       0.421659
concavity_se                         0.439167       0.342627
concave points_se                    0.615634       0.393298
symmetry_se                          0.095351       0.449137
fractal_dimension_se                 0.257584       0.331786
radius_worst                         0.830318       0.185728
texture_worst                        0.292752       0.090651
perimeter_worst                      0.855923       0.219169
area_worst                           0.809630       0.177193
smoothness_worst                     0.452753       0.426675
compactness_worst                    0.667454       0.473200
concavity_worst                      0.752399       0.433721
concave points_worst                 0.910155       0.430297
```

```
symmetry_worst                       0.375744       0.699826
fractal_dimension_worst              0.368661       0.438413
diagnosis                           -0.776614      -0.330499

                          fractal_dimension_mean  ...  texture_worst  \
radius_mean                             -0.311631  ...       0.297008
texture_mean                            -0.076437  ...       0.912045
perimeter_mean                          -0.261477  ...       0.303038
area_mean                               -0.283110  ...       0.287489
smoothness_mean                          0.584792  ...       0.036072
compactness_mean                         0.565369  ...       0.248133
concavity_mean                           0.336783  ...       0.299879
concave points_mean                      0.166917  ...       0.292752
symmetry_mean                            0.479921  ...       0.090651
fractal_dimension_mean                   1.000000  ...      -0.051269
radius_se                                0.000111  ...       0.194799
texture_se                               0.164174  ...       0.409003
perimeter_se                             0.039830  ...       0.200371
area_se                                 -0.090170  ...       0.196497
smoothness_se                            0.401964  ...      -0.074743
compactness_se                           0.559837  ...       0.143003
concavity_se                             0.446630  ...       0.100241
concave points_se                        0.341198  ...       0.086741
symmetry_se                              0.345007  ...      -0.077473
fractal_dimension_se                     0.688132  ...      -0.003195
radius_worst                            -0.253691  ...       0.359921
texture_worst                           -0.051269  ...       1.000000
perimeter_worst                         -0.205151  ...       0.365098
area_worst                              -0.231854  ...       0.345842
smoothness_worst                         0.504942  ...       0.225429
compactness_worst                        0.458798  ...       0.360832
concavity_worst                          0.346234  ...       0.368366
concave points_worst                     0.175325  ...       0.359755
symmetry_worst                           0.334019  ...       0.233027
fractal_dimension_worst                  0.767297  ...       0.219122
diagnosis                                0.012838  ...      -0.456903

                          perimeter_worst  area_worst  smoothness_worst
\
radius_mean                      0.965137    0.941082          0.119616

texture_mean                     0.358040    0.343546          0.077503

perimeter_mean                   0.970387    0.941550          0.150549

area_mean                        0.959120    0.959213          0.123523

smoothness_mean                  0.238853    0.206718          0.805324

compactness_mean                 0.590210    0.509604          0.565541
```

concavity_mean	0.729565	0.675987	0.448822
concave points_mean	0.855923	0.809630	0.452753
symmetry_mean	0.219169	0.177193	0.426675
fractal_dimension_mean	-0.205151	-0.231854	0.504942
radius_se	0.719684	0.751548	0.141919
texture_se	-0.102242	-0.083195	-0.073658
perimeter_se	0.721031	0.730713	0.130054
area_se	0.761213	0.811408	0.125389
smoothness_se	-0.217304	-0.182195	0.314457
compactness_se	0.260516	0.199371	0.227394
concavity_se	0.226680	0.188353	0.168481
concave points_se	0.394999	0.342271	0.215351
symmetry_se	-0.103753	-0.110343	-0.012662
fractal_dimension_se	-0.001000	-0.022736	0.170568
radius_worst	0.993708	0.984015	0.216574
texture_worst	0.365098	0.345842	0.225429
perimeter_worst	1.000000	0.977578	0.236775
area_worst	0.977578	1.000000	0.209145
smoothness_worst	0.236775	0.209145	1.000000
compactness_worst	0.529408	0.438296	0.568187
concavity_worst	0.618344	0.543331	0.518523
concave points_worst	0.816322	0.747419	0.547691
symmetry_worst	0.269493	0.209146	0.493838
fractal_dimension_worst	0.138957	0.079647	0.617624
diagnosis	-0.782914	-0.733825	-0.421465

	compactness_worst	concavity_worst	\
radius_mean	0.413463	0.526911	
texture_mean	0.277830	0.301025	
perimeter_mean	0.455774	0.563879	
area_mean	0.390410	0.512606	
smoothness_mean	0.472468	0.434926	
compactness_mean	0.865809	0.816275	
concavity_mean	0.754968	0.884103	
concave points_mean	0.667454	0.752399	
symmetry_mean	0.473200	0.433721	
fractal_dimension_mean	0.458798	0.346234	
radius_se	0.287103	0.380585	
texture_se	-0.092439	-0.068956	
perimeter_se	0.341919	0.418899	
area_se	0.283257	0.385100	
smoothness_se	-0.055558	-0.058298	
compactness_se	0.678780	0.639147	
concavity_se	0.484858	0.662564	
concave points_se	0.452888	0.549592	
symmetry_se	0.060255	0.037119	
fractal_dimension_se	0.390159	0.379975	
radius_worst	0.475820	0.573975	
texture_worst	0.360832	0.368366	
perimeter_worst	0.529408	0.618344	
area_worst	0.438296	0.543331	
smoothness_worst	0.568187	0.518523	
compactness_worst	1.000000	0.892261	
concavity_worst	0.892261	1.000000	
concave points_worst	0.801080	0.855434	
symmetry_worst	0.614441	0.532520	
fractal_dimension_worst	0.810455	0.686511	
diagnosis	-0.590998	-0.659610	

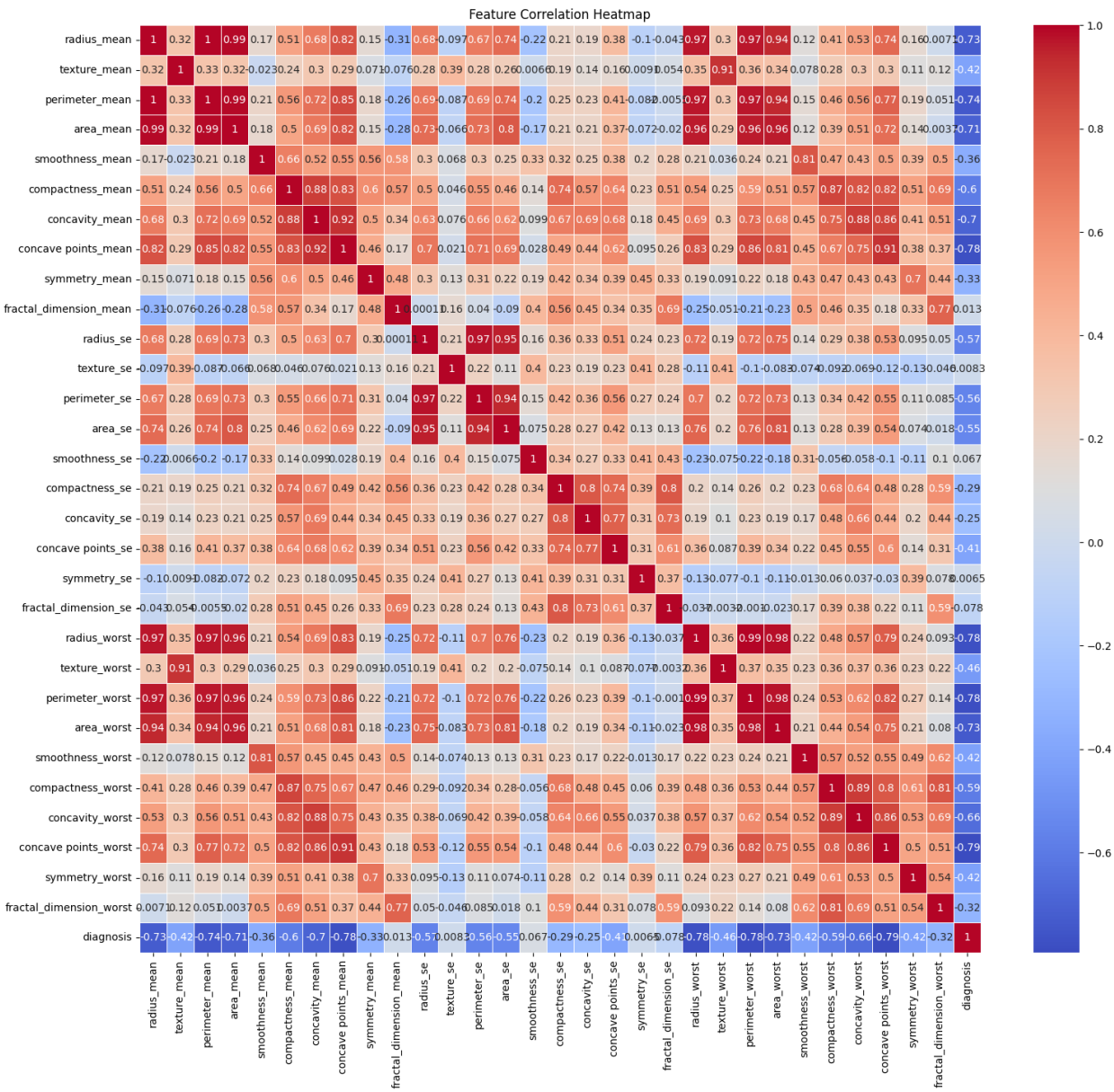
	concave points_worst	symmetry_worst	\
radius_mean	0.744214	0.163953	
texture_mean	0.295316	0.105008	
perimeter_mean	0.771241	0.189115	
area_mean	0.722017	0.143570	
smoothness_mean	0.503053	0.394309	
compactness_mean	0.815573	0.510223	
concavity_mean	0.861323	0.409464	
concave points_mean	0.910155	0.375744	
symmetry_mean	0.430297	0.699826	
fractal_dimension_mean	0.175325	0.334019	
radius_se	0.531062	0.094543	
texture_se	-0.119638	-0.128215	
perimeter_se	0.554897	0.109930	
area_se	0.538166	0.074126	
smoothness_se	-0.102007	-0.107342	

compactness_se	0.483208	0.277878
concavity_se	0.440472	0.197788
concave points_se	0.602450	0.143116
symmetry_se	-0.030413	0.389402
fractal_dimension_se	0.215204	0.111094
radius_worst	0.787424	0.243529
texture_worst	0.359755	0.233027
perimeter_worst	0.816322	0.269493
area_worst	0.747419	0.209146
smoothness_worst	0.547691	0.493838
compactness_worst	0.801080	0.614441
concavity_worst	0.855434	0.532520
concave points_worst	1.000000	0.502528
symmetry_worst	0.502528	1.000000
fractal_dimension_worst	0.511114	0.537848
diagnosis	-0.793566	-0.416294

	fractal_dimension_worst	diagnosis
radius_mean	0.007066	-0.730029
texture_mean	0.119205	-0.415185
perimeter_mean	0.051019	-0.742636
area_mean	0.003738	-0.708984
smoothness_mean	0.499316	-0.358560
compactness_mean	0.687382	-0.596534
concavity_mean	0.514930	-0.696360
concave points_mean	0.368661	-0.776614
symmetry_mean	0.438413	-0.330499
fractal_dimension_mean	0.767297	0.012838
radius_se	0.049559	-0.567134
texture_se	-0.045655	0.008303
perimeter_se	0.085433	-0.556141
area_se	0.017539	-0.548236
smoothness_se	0.101480	0.067016
compactness_se	0.590973	-0.292999
concavity_se	0.439329	-0.253730
concave points_se	0.310655	-0.408042
symmetry_se	0.078079	0.006522
fractal_dimension_se	0.591328	-0.077972
radius_worst	0.093492	-0.776454
texture_worst	0.219122	-0.456903
perimeter_worst	0.138957	-0.782914
area_worst	0.079647	-0.733825
smoothness_worst	0.617624	-0.421465
compactness_worst	0.810455	-0.590998
concavity_worst	0.686511	-0.659610
concave points_worst	0.511114	-0.793566
symmetry_worst	0.537848	-0.416294
fractal_dimension_worst	1.000000	-0.323872
diagnosis	-0.323872	1.000000

```
[31 rows x 31 columns]
```

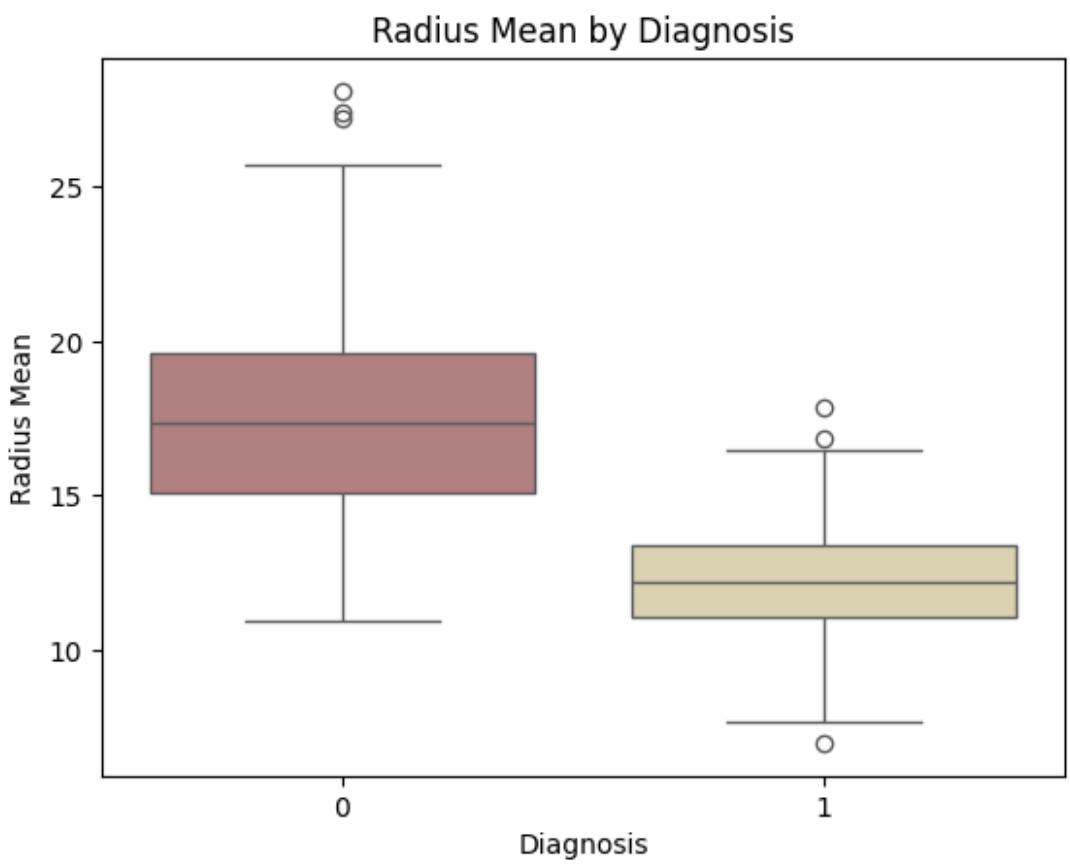
```
#Heatmap
plt.figure(figsize=(18, 16))
sns.heatmap(df.corr(), annot=True, cmap='coolwarm', linewidths=0.5)
plt.title('Feature Correlation Heatmap')
plt.show()
```




```
#Box-Plot
sns.boxplot(x='diagnosis', y='radius_mean', data=df,palette='pink')
plt.title('Radius Mean by Diagnosis')
plt.xlabel('Diagnosis')
```

```
plt.ylabel('Radius Mean')
plt.show()
```

```
C:\Users\LENOVO\AppData\Local\Temp\ipykernel_9776\410977824.py:2:
FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be
removed in v0.14.0. Assign the `x` variable to `hue` and set
`legend=False` for the same effect.

  sns.boxplot(x='diagnosis', y='radius_mean', data=df,palette='pink')
```




```
X=df.iloc[:,:-1]
y=df['diagnosis']

print(X)
```

```
     radius_mean  texture_mean  perimeter_mean  area_mean
smoothness_mean  \
0          17.99         10.38          122.80     1001.0
0.11840
1          20.57         17.77          132.90     1326.0
0.08474
```

```
2          19.69         21.25          130.00     1203.0
0.10960
3          11.42         20.38           77.58      386.1
0.14250
4          20.29         14.34          135.10     1297.0
0.10030
..           ...           ...             ...        ...
...
564        21.56         22.39          142.00     1479.0
0.11100
565        20.13         28.25          131.20     1261.0
0.09780
566        16.60         28.08          108.30      858.1
0.08455
567        20.60         29.33          140.10     1265.0
0.11780
568         7.76         24.54           47.92      181.0
0.05263

     compactness_mean  concavity_mean  concave points_mean
symmetry_mean  \
0             0.27760         0.30010              0.14710
0.2419
1             0.07864         0.08690              0.07017
0.1812
2             0.15990         0.19740              0.12790
0.2069
3             0.28390         0.24140              0.10520
0.2597
4             0.13280         0.19800              0.10430
0.1809
..                ...             ...                  ...
...
564           0.11590         0.24390              0.13890
0.1726
565           0.10340         0.14400              0.09791
0.1752
566           0.10230         0.09251              0.05302
0.1590
567           0.27700         0.35140              0.15200
0.2397
568           0.04362         0.00000              0.00000
0.1587

     fractal_dimension_mean  ...  radius_worst  texture_worst  \
0                   0.07871  ...        25.380          17.33
1                   0.05667  ...        24.990          23.41
2                   0.05999  ...        23.570          25.53
3                   0.09744  ...        14.910          26.50
```

```
4                   0.05883  ...        22.540          16.67
..                      ...  ...           ...            ...
564                 0.05623  ...        25.450          26.40
565                 0.05533  ...        23.690          38.25
566                 0.05648  ...        18.980          34.12
567                 0.07016  ...        25.740          39.42
568                 0.05884  ...         9.456          30.37

     perimeter_worst  area_worst  smoothness_worst  compactness_worst
\
0             184.60      2019.0           0.16220            0.66560

1             158.80      1956.0           0.12380            0.18660

2             152.50      1709.0           0.14440            0.42450

3              98.87       567.7           0.20980            0.86630

4             152.20      1575.0           0.13740            0.20500

..               ...         ...               ...                ...

564           166.10      2027.0           0.14100            0.21130

565           155.00      1731.0           0.11660            0.19220

566           126.70      1124.0           0.11390            0.30940

567           184.60      1821.0           0.16500            0.86810

568            59.16       268.6           0.08996            0.06444


     concavity_worst  concave points_worst  symmetry_worst  \
0             0.7119                0.2654          0.4601
1             0.2416                0.1860          0.2750
2             0.4504                0.2430          0.3613
3             0.6869                0.2575          0.6638
4             0.4000                0.1625          0.2364
..               ...                   ...             ...
564           0.4107                0.2216          0.2060
565           0.3215                0.1628          0.2572
566           0.3403                0.1418          0.2218
567           0.9387                0.2650          0.4087
568           0.0000                0.0000          0.2871

     fractal_dimension_worst
0                    0.11890
1                    0.08902
2                    0.08758
3                    0.17300
```

```
4                     0.07678
..                        ...
564                   0.07115
565                   0.06637
566                   0.07820
567                   0.12400
568                   0.07039

[569 rows x 30 columns]
```

```
print(y)
```

```
0      0
1      0
2      0
3      0
4      0
      ..
564    0
565    0
566    0
567    0
568    1
Name: diagnosis, Length: 569, dtype: int64
```

```
from sklearn.preprocessing import StandardScaler

 scaler=StandardScaler()
 ss=scaler.fit_transform(X)

from sklearn.model_selection import train_test_split

X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2,rando
m_state=42)

 print(X.shape,X_train.shape,X_test.shape)
```

```
(569, 30) (455, 30) (114, 30)
```

```
 print(y.shape,y_train.shape,y_test.shape)
```

```
(569,) (455,) (114,)
```

```
from sklearn.linear_model import LogisticRegression
from sklearn.neighbors import KNeighborsClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.svm import SVC
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import
accuracy_score,precision_score,recall_score,f1_score
```

```
models = {
    "Logistic Regression": LogisticRegression(max_iter=2000,
solver='liblinear'),
    "KNN": KNeighborsClassifier(n_neighbors=5),
    "Decision Tree": DecisionTreeClassifier(random_state=42),
    "Random Forest": RandomForestClassifier(random_state=42),
    "SVM": SVC(kernel='linear'),
    "Naive Bayes": GaussianNB()
}
```

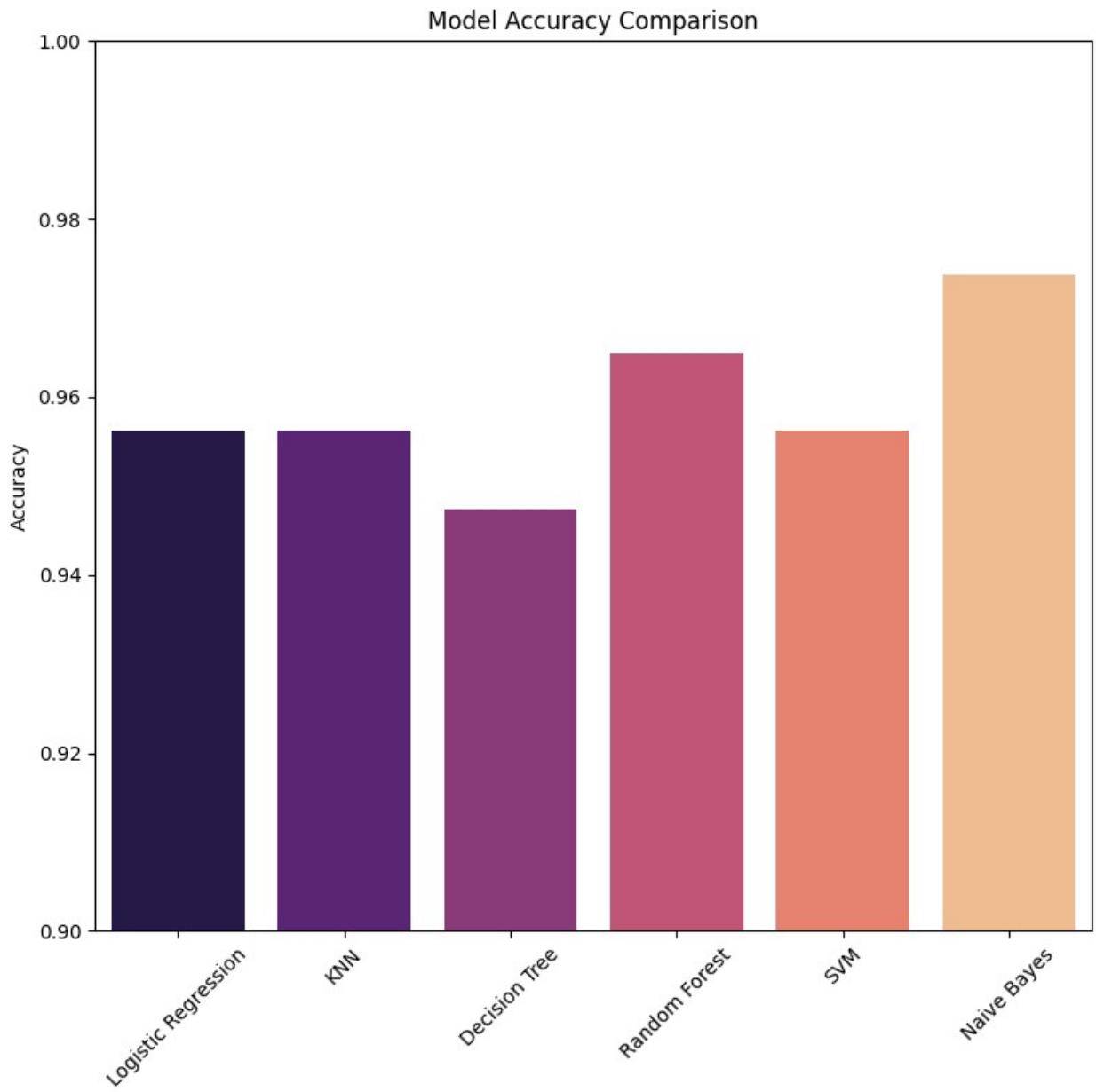
```
models = {
    "Logistic Regression":
LogisticRegression(max_iter=2000,solver='liblinear'),
    "KNN": KNeighborsClassifier(n_neighbors=5),
    "Decision Tree": DecisionTreeClassifier(random_state=42),
    "Random Forest": RandomForestClassifier(random_state=42),
    "SVM": SVC(kernel='linear'),
    "Naive Bayes": GaussianNB()
 }

 for name, model in models.items():
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    accuracy = accuracy_score(y_test, y_pred)
    precision = precision_score(y_test, y_pred, average='weighted')
    recall = recall_score(y_test, y_pred, average='weighted')
    f1 = f1_score(y_test, y_pred, average='weighted')

    print(f"--- {name} ---")
    print(f"Accuracy : {accuracy:.4f}")
    print(f"Precision: {precision:.4f}")
    print(f"Recall   : {recall:.4f}")
    print(f"F1 Score : {f1:.4f}")
    print("\n")
```

```
--- Logistic Regression ---
Accuracy : 0.9561
Precision: 0.9569
Recall   : 0.9561
F1 Score : 0.9558


--- KNN ---
Accuracy : 0.9561
Precision: 0.9590
Recall   : 0.9561
F1 Score : 0.9555


--- Decision Tree ---
```

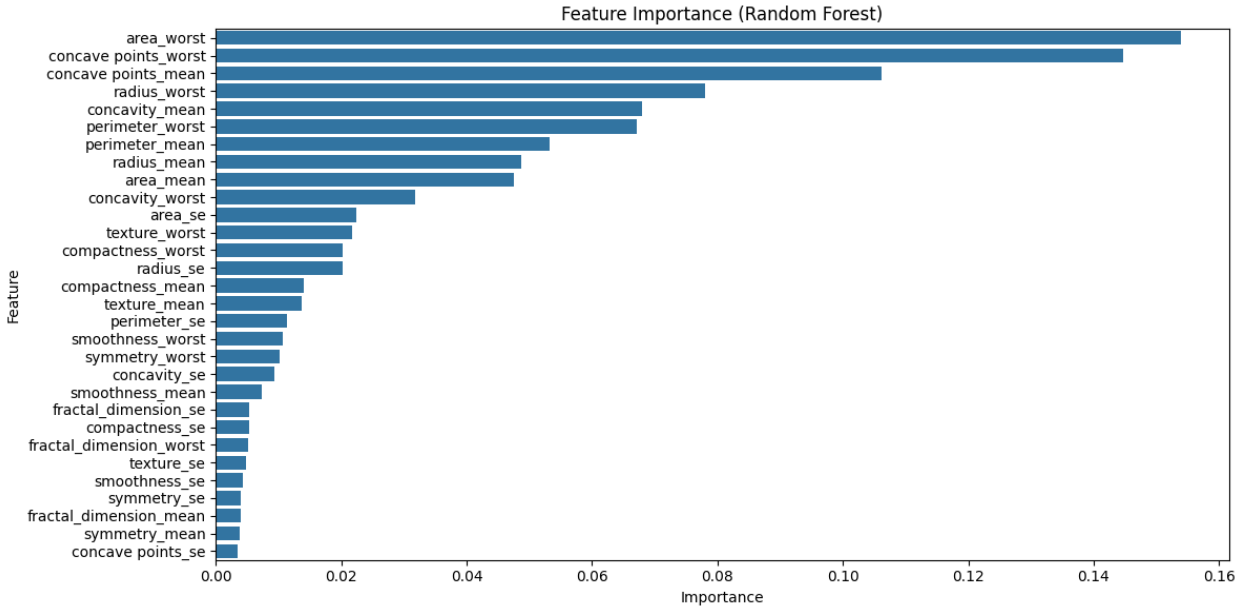
```
Accuracy : 0.9474
Precision: 0.9474
Recall   : 0.9474
F1 Score : 0.9474


--- Random Forest ---
Accuracy : 0.9649
Precision: 0.9652
Recall   : 0.9649
F1 Score : 0.9647


--- SVM ---
Accuracy : 0.9561
Precision: 0.9569
Recall   : 0.9561
F1 Score : 0.9558


--- Naive Bayes ---
Accuracy : 0.9737
Precision: 0.9748
Recall   : 0.9737
F1 Score : 0.9735
```

```
plt.figure(figsize=(8,8))
sns.barplot(x=list(results.keys()),
y=list(results.values()),palette='magma')
plt.title('Model Accuracy Comparison')
plt.ylabel('Accuracy')
plt.xticks(rotation=45)
plt.ylim(0.9, 1.0)
plt.tight_layout()
plt.show()
```

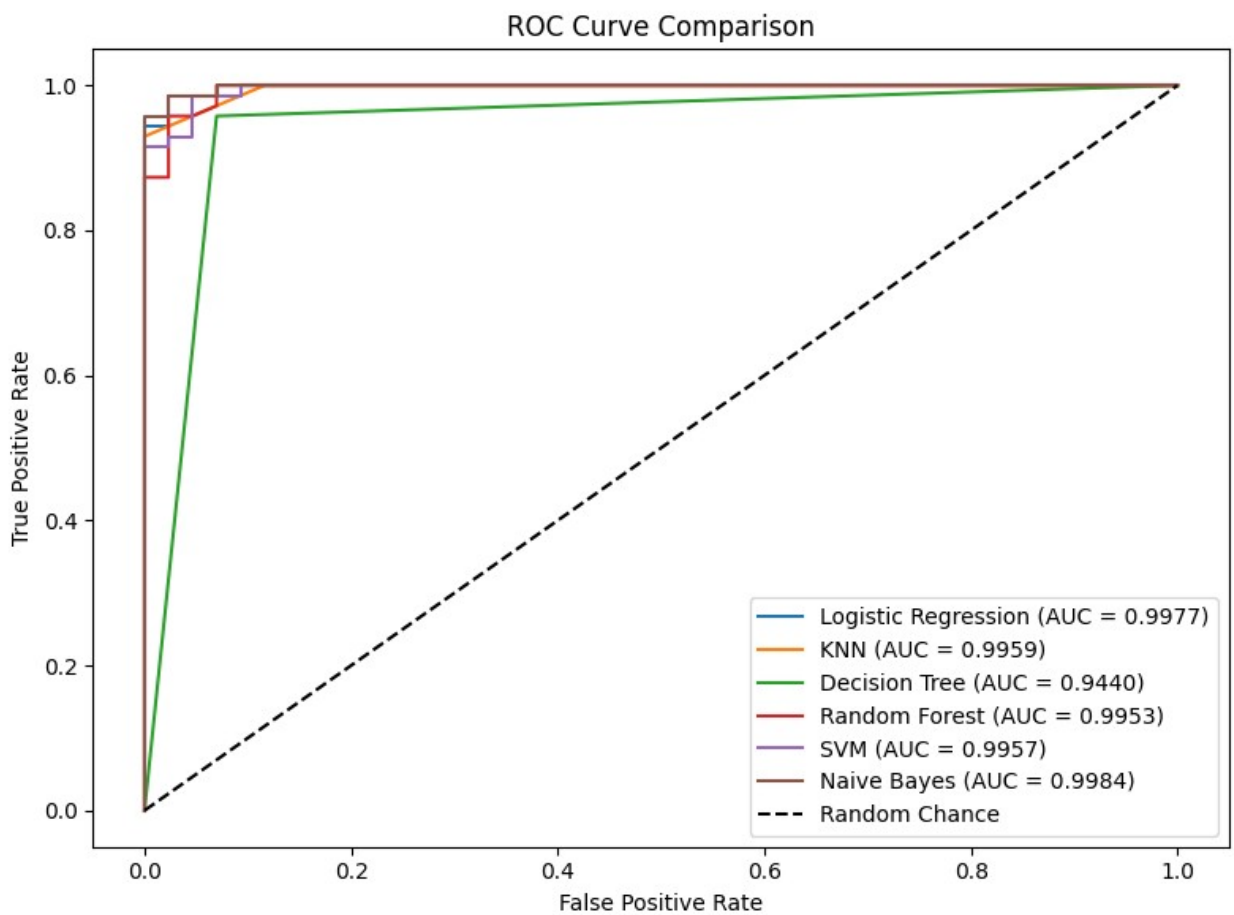
```
C:\Users\LENOVO\AppData\Local\Temp\ipykernel_9776\544123435.py:2:
FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be
removed in v0.14.0. Assign the `x` variable to `hue` and set
`legend=False` for the same effect.

  sns.barplot(x=list(results.keys()),
y=list(results.values()),palette='magma')
```




```
importances = models["Random Forest"].feature_importances_
feature_names = X.columns
indices = np.argsort(importances)[::-1]

plt.figure(figsize=(12, 6))
sns.barplot(x=importances[indices], y=feature_names[indices])
plt.title('Feature Importance (Random Forest)')
plt.xlabel('Importance')
plt.ylabel('Feature')
plt.tight_layout()
plt.show()
```




```python
from sklearn.metrics import roc_auc_score, roc_curve

plt.figure(figsize=(8, 6))

for name, model in models.items():
    model.fit(X_train, y_train)

    # Check if model supports probability or decision scores
    if hasattr(model, "predict_proba"):
        y_scores = model.predict_proba(X_test)[:, 1]  # probability
for positive class
    elif hasattr(model, "decision_function"):
        y_scores = model.decision_function(X_test)  # decision
function scores
    else:
        print(f"{name} does not support probability estimates or
decision function.")
        continue

    # Calculate AUC
    auc = roc_auc_score(y_test, y_scores)
    print(f"{name} AUC: {auc:.4f}")

    # Compute ROC curve
    fpr, tpr, thresholds = roc_curve(y_test, y_scores)

    # Plot ROC curve
    plt.plot(fpr, tpr, label=f"{name} (AUC = {auc:.4f})")

# Plot the random chance line
plt.plot([0, 1], [0, 1], 'k--', label="Random Chance")
```

```
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('ROC Curve Comparison')
plt.legend(loc='lower right')
plt.tight_layout()
plt.show()
```

```
Logistic Regression AUC: 0.9977
KNN AUC: 0.9959
Decision Tree AUC: 0.9440
Random Forest AUC: 0.9953
SVM AUC: 0.9957
Naive Bayes AUC: 0.9984
```